Thrifting the Transition: How Energy Policy Reshaped the Silver-Solar Price Channel

1 Introduction

Silver occupies a unique position in global commodity markets. While it has traditionally been viewed as a monetary metal and safe-haven asset, it is increasingly being used as an industrial input (MMTC PAMP, 2026). Its demand profile is being reshaped by the global energy transition (Oxford Economics, 2025). At current technology levels, there is no viable substitute for silver paste in cell conductivity. As a result, solar deployment has accelerated, driven by falling panel costs and policy support (Kishore, Kumar and Ippili, 2025). Whether this shift in demand has translated into silver prices remains an open empirical question.

This paper tests whether major energy policy commitments in 2015 and 2016 (the Paris Agreement, China's 13th Five-Year Plan solar targets, and the US Investment Tax Credit extension) created a structural break in the relationship between silver prices and solar activity. These simultaneous commitments provide a clear and theoretically motivated break date. In the paper, two hypotheses are tested:

H1: Silver prices became more responsive to global industrial production after 2016
H2: The solar PV channel drives a structural change in silver prices after January 2016

The results support H2 but not H1. A Chow test identifies a significant structural break in January 2016 ($F(6,175) = 6.19$, $p < 0.001$). The industrial production interaction is insignificant, suggesting the break is not due to a general shift in silver's industrial role. In contrast, the solar interaction term is significant at the 1% level, indicating a narrow, solar-specific pricing channel rather than a broad industrialisation. The failure of H1, therefore, strengthens rather than undermines H2 by ruling out a general effect.

After 2016, the solar coefficient reverses sign, consistent with a technology efficiency channel. As solar deployment accelerated, manufacturers reduced silver intensity per panel from 400mg per cell in 2010 to under 100mg by the early 2020s (ITRPV, 2023).

This paper contributes to the commodity price literature (Frankel, 2008; Batten, Ciner and Lucey, 2010), the critical minerals literature (IEA, 2021), and the structural break literature in

commodity economics (Perron, 1989). The remainder of the paper reviews the literature, outlines the empirical approach, describes the data, presents the results and discusses robustness checks.

2 Literature Review

The determinants of silver have traditionally been studied within the broader context of precious metal markets. Silver is often grouped with gold as a safe-haven asset and inflation hedge. However, it responds differently to macroeconomic conditions with distinct drivers of volatility (Dinh et al., 2022). This motivates treating silver as a separate asset rather than a proxy for gold. Real interest rates and the dollar index are also key drivers of commodity prices (Akram, 2009), motivating their inclusion as controls. In this paper, gold, treasury yields and the dollar index are consistently significant across all model specifications.

More recent literature focuses on critical minerals and the energy transition. Silver is a key input into photovoltaic technology (Rout et al., 2025). Under accelerated transition scenarios, clean energy could account for over 40% of silver demand by 2040 (BELLINI, 2025). Growth in solar applications has already increased industrial demand for silver, coinciding with significant technological changes in how it is used (Cattaneo et al., 2025). Literature highlights a "*thrifting*" trend where silver usage per panel declines over the years (Zhang et al., 2023). This is central to interpreting the sign reversal in the post-2016 interaction as it points to an efficiency effect rather than a simple demand-driven relationship.

The empirical approach comes from the structural break literature. In commodity markets, rolling regression methods show that the relationship between macroeconomic variables and metal prices can change over time. (Kucher and McCoskey, 2016). This variation is consistent with the results presented in this paper. More broadly, policy-driven demand shocks have been shown to generate structural breaks in commodity price dynamics (Allayioti and Venditti, 2024). However, whether the 2015-2016 energy policy commitments created such a break for silver remains an open question. This paper addresses that gap directly.

3 Empirical Approach

This paper uses a structural break regression approach to test whether the relationship between silver prices and solar PV capacity changed after January 2016. The structural break

is identified with a Chow test, using an interaction term regression allowing all slope coefficients to differ between the pre-2016 and post-2016 periods. This approach tests whether there are significant changes in parameters while assuming a common error variance across the full sample.

The dependent variable is the monthly log return on the real silver price (Δlog silver). This choice ensures stationarity, confirmed by Augmented Dickey-Fuller unit root tests. All level variables are integrated of order one and enter the model in first differences, except for solar capacity, which is integrated of order two and enters as a second difference (Δ^2solar).

Three models are estimated sequentially:

Model 1 asks the simplest question: Is there any overall shift in silver returns after 2016, after accounting for standard precious metals and macroeconomic controls? If the post-2016 dummy is insignificant, it suggests that any structural change is not just an intercept shift but occurs through a specific channel.

$$\Delta\log(silver)_t = \alpha + \beta_1 Post2016_t + \beta_2 \Delta\log(gold)_t + \beta_3 \Delta treasury_t + \beta_4 \Delta dollar_t + \varepsilon_t$$

The post-2016 equals 1 from January 2016 onwards and 0 otherwise, capturing the combined effect of the Paris Agreement, China's 13th Five-Year Plan, and the US Investment Tax Credit extension as a single exogenous treatment.

Model 2 tests whether the structural break could be explained by broad industrialisation. It adds global industrial production and its interaction with the post-2016 dummy:

$$\begin{aligned}\Delta log(silver)_t = \alpha &+ \beta_1 \Delta log(indpro)_t + \beta_2 Post2016_t \\ &+ \beta_3(\Delta log(indpro)_t \times Post2016_t) + \beta_4 \Delta log(gold)_t + \beta_5 \Delta treasury_t \\ &+ \beta_6 \Delta dollar_t + \varepsilon_t\end{aligned}$$

H1 is tested by the significance of β_3 i.e., whether the industrial production drives structural change.

Model 3 narrows the equation further by isolating the energy transition as a pricing mechanism. It adds solar PV channel and interaction with the post-2016 dummy:

Model 3 tests H2 by adding solar capacity and its post-2016 interaction:

$$\Delta log(silver)_t = \alpha + \beta_1 \Delta log(indpro)_t + \beta_2 \Delta^2 solar_t + \beta_3 Post2016_t + \beta_4(\Delta log(indpro)_t \times Post2016_t) + \beta_5(\Delta^2 solar_t \times Post2016_t) + \beta_6 \Delta log(gold)_t + \beta_7 \Delta treasury_t + \beta_8 \Delta dollar_t + \varepsilon_t$$

H2 by the significance of β_5 in Model 3. A joint Chow test of all post-2016 interaction terms tests whether the relationship between silver returns and explanatory variables changed after 2016.

Gold is included as a precious metal co-movement control, with an expected positive coefficient. Treasury yields capture the opportunity cost of holding non-yielding assets, expected to be positive given silver's inflation-hedge properties. The dollar index captures currency effects on dollar-denominated commodity prices, expected to be negative. A COVID-19 dummy controlling for March through December 2020 is included in robustness checks. All models are estimated with heteroskedasticity-robust standard errors given the financial time series context.

4 Data Analysis

The dataset includes 192 monthly observations from January 2010 to December 2025, drawn from five primary sources. Monthly averages of silver and gold prices (LBMA, 2026) are deflated by the US Consumer Price Index to obtain real prices. The CPB World Trade Monitor provides a monthly index of global industrial output while the 10-year US treasury yield (FRED, 2025) and trade-weighted dollar index (FRED, 2006) serve as macroeconomic controls. Cumulative global solar photovoltaic capacity (IRENA, 2025) captures the energy transition channel. To control for pandemic-related market disruption, a COVID-19 dummy takes the value of 1 for March through December 2020.

The sample period begins in January 2010, capturing the first wave of utility-scale solar deployment and ends in December 2025, providing 72 pre-break and 120 post-break observations. All price series are converted to monthly log returns to ensure stationarity. Unit root tests confirm that all variables are integrated of order one (I (1)) and stationary in first differences, except solar capacity, which is integrated of order two (I(2)). This is confirmed by

both Augmented Dickey-Fuller tests across all standard lag specifications, with the results presented in Table 1.

Table 1: Augmented Dickey Fuller Test Results

Variable	ADF Level p-value	ADF Diff p-value	Order
log_silver	0.970	0.033	I(1)
log_gold	0.987	0.000	I(1)
log_indpro	0.230	0.000	I(1)
treasury_10y	0.365	0.024	I(1)
dollar_index	0.553	0.001	I(1)
solar_mw	0.575	0.258	I(2)
d_solar_mw	—	0.003	I(2)→I(1)

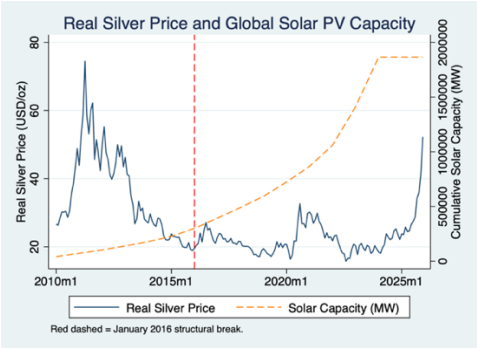
Table 2 shows the summary statistics for the key variables. The mean monthly silver log return is close to zero at 0.001, with a standard deviation of 0.088, reflecting the high volatility characteristic of precious metal markets.

Table 2: Summary Statistics

	(1)				
	mean	sd	min	max	count
dlog_silver	0.0035	0.0892	-0.3098	0.2528	191
dlog_gold	0.0051	0.0450	-0.1611	0.1204	191
dlog_indpro	0.0020	0.0096	-0.0886	0.0455	188
d_solar_mw	9550.2150	10174.9959	0.0000	37661.8047	191
d_treasury	0.0021	0.1994	-0.7000	0.6212	191
d_dollar	0.1453	1.3422	-2.9384	4.3122	191
post2016	0.6250	0.4854	0.0000	1.0000	192
covid	0.0521	0.2228	0.0000	1.0000	192
N	192				

Figure 1 illustrates the divergence between real silver prices and cumulative solar capacity over the sample period. Silver prices peaked in 2011 and trended downward through the 2016 break date, while solar capacity has grown exponentially, with the steepest acceleration occurring after January 2016.

Figure 1: Real Silver Price and Global Solar PV Capacity



5 Results

A joint Chow test of all post-2016 interaction terms yields $F(6,175) = 6.19$ ($p < 0.001$), rejecting the null hypothesis of parameter stability. This confirms a statistically significant structural break in January 2016. Alternative break dates of 2015 or 2017 are tested and produce inferior model fit, supporting the choice of January 2016 as theoretically motivated by the simultaneous policy commitments during this period.

Table 3 presents the three model specifications. Model 1 establishes that the post-2016 dummy alone is insignificant ($p = 0.311$), indicating no straightforward intercept shift in silver returns. The control variables behave as expected: gold returns are strongly positive and significant throughout ($\beta \approx 1.53$, $p < 0.001$), confirming silver's co-movement with precious metal markets. Treasury yields enter positively and significantly ($\beta \approx 0.088$, $p < 0.001$) and the dollar index enters negatively ($\beta \approx -0.010$, $p < 0.01$). The stability of these coefficients across all three models provides confidence in the control structure.

Table 3: Main Regression Results

	(1) Model 1	(2) Model 2	(3) Model 3
Post-2016 Dummy	-0.0084	-0.0172**	-0.0166*
	(0.0083)	(0.0087)	(0.0088)
Gold Price (Δlog)	1.5264***	1.5198***	1.5266***
	(0.1029)	(0.1027)	(0.1037)
10Y Treasury (Δ)	0.0884***	0.0853***	0.0851***
	(0.0240)	(0.0240)	(0.0241)
Dollar Index (Δ)	-0.0098***	-0.0091***	-0.0090***
	(0.0031)	(0.0031)	(0.0031)
Industrial Production (Δlog)		-2.5461	-2.5810

		(2.1914)	(2.1954)
IndProd × Post-2016		2.8459	2.8863
		(2.2194)	(2.2220)
Solar Capacity (Δ^2MW)			0.0000***
			(0.0000)
Solar × Post-2016			-0.0000***
			(0.0000)
Constant	0.0022	0.0082	0.0076
	(0.0064)	(0.0072)	(0.0073)
N	191	188	187
R-squared	0.6135	0.6279	0.6308

Robust standard errors in parentheses.
* p<0.10 ** p<0.05 *** p<0.01
All variables in first differences. Solar in second differences (I(2)).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Model 2 tests H1 by adding global industrial production and its post-2016 interaction. The interaction term (IndProd × Post-2016) is insignificant (p = 0.201). H1 is therefore not supported. This indicates that silver did not become more responsive to broad industrial activity after 2016, ruling out the general industrial channel as the driver of the structural break.

Model 3 tests H2 by adding solar capacity and its post-2016 interaction. The solar interaction term (Solar × Post-2016) is highly significant at the 1% level (p = 0.001), and the joint test of both interaction terms yields F(2,178) = 5.57 (p = 0.005), confirming H2. The solar channel specifically drives a structural change in silver prices after January 2016.

The result is striking. Before January 2016, increases in solar capacity were positively associated with silver returns (β = 0.0000252, p = 0.001), reflecting a straightforward demand channel. After the break, this relationship reverses (β = -0.0000252, p = 0.001), so higher solar capacity is now linked to lower silver returns. This sharp reversal occurs precisely at the break date, as shown in the rolling 36-month regressions in Figures 2 and 3.

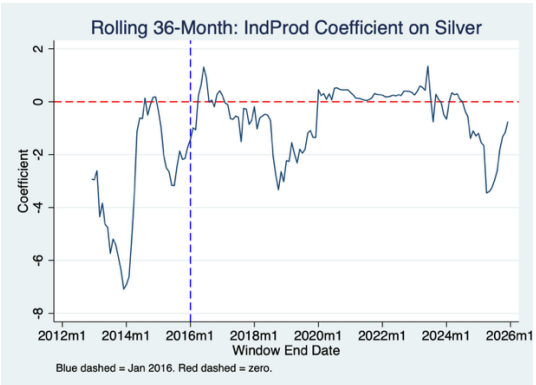


Figure 2

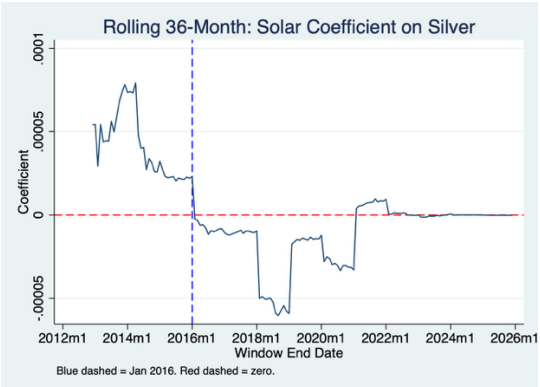


Figure 3

While the near-perfect symmetry of the pre- and post-2016 coefficients partly reflects the construction of the interaction terms, multicollinearity is high (VIF ≈ 365), so the exact magnitude should be interpreted cautiously. Nevertheless, the sign reversal itself is meaningful**.** It aligns with the technology efficiency channel (VDMA, 2024): post-2016 policy commitments accelerated solar deployment while encouraging manufacturers to reduce silver use per panel. This "*thrifting*" dynamic weakened the previous price-transmission effect, meaning that larger increases in solar capacity now reduce silver price pressure rather than increase it.

6 Robustness Checks

Model 3 passes all standard OLS diagnostic tests. The Breusch-Godfrey test finds no evidence of serial correlation across lag lengths of 1, 2, 3, 6, and 12 months (all $p > 0.05$. The Breusch-Pagan test finds no significant heteroskedasticity ($\chi^2(1) = 2.65$, $p = 0.103$), though robust standard errors are retained throughout as a precaution for financial time series. The ARCH test detects mild volatility clustering at lags 1 and 3 ($p = 0.022$ and $p = 0.015$). This is consistent with the well-documented volatility clustering in commodity markets (Brockwell, 2025). This highlights a limitation of the OLS framework employed and suggests a GARCH extension for future research. Residuals are approximately normally distributed as confirmed visually in Figure 4 and supported by the skewness test ($p = 0.927$), with mild excess kurtosis ($p = 0.001$) typical of financial return series. Figure 5 confirms that residuals are well-behaved and mostly stationary over time with no systematic patterns around the break date.

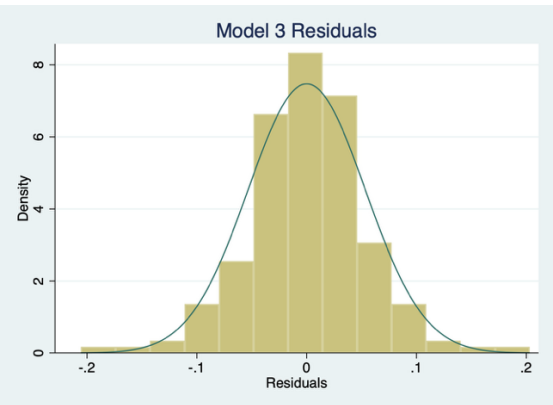


Figure 4

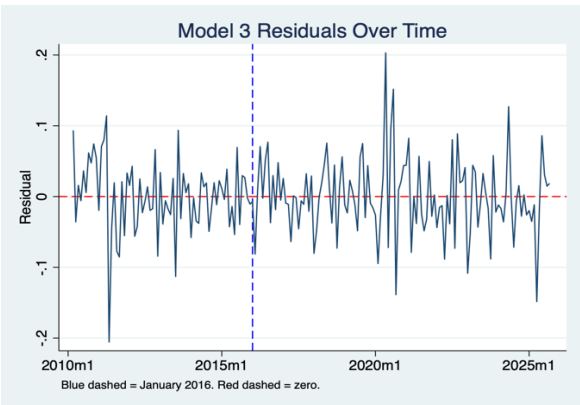


Figure 5

Variance inflation factors are high for the solar interaction pair (VIF = 365), which is expected when interaction terms are included and does not indicate problematic multicollinearity in the underlying data. All base control variables have VIF below 1.25.

Four robustness checks are reported in Table 4. First, using a six-month lag for solar capacity preserves the sign and direction of the solar interaction (p = 0.066), consistent with procurement cycle dynamics in solar manufacturing (Novarr, 2023). Second, a copper falsification test substitutes copper returns as the dependent variable. The results show that the solar interaction is entirely insignificant for copper (p = 0.745), confirming that the effect is silver-specific and not a broad commodity phenomenon. Third, excluding the ten COVID-19-affected months leaves the solar interaction unchanged (p = 0.003). This shows that the results are not driven by pandemic-related disruptions. Fourth, alternative break dates of 2015 and 2017 produce inferior model fit compared to 2016, supporting the theoretical motivation for the January 2016 break date.

Table 4: Robustness Checks

	(1) Baseline	(2) 6M Lag Solar	(3) Copper Falsif.	(4) No COVID
dlog_indpro	-2.5810	-4.0050*	3.2441***	-2.5287
	(2.1954)	(2.3546)	(1.2351)	(2.2368)
d2_solar_mw	0.0000***		0.0000	0.0000***
	(0.0000)		(0.0000)	(0.0000)
post2016	-0.0166*	-0.0207**	0.0109	-0.0189**
	(0.0088)	(0.0090)	(0.0073)	(0.0089)
indprod_post	2.8863	4.3260*	-2.9681**	3.0106
	(2.2220)	(2.3720)	(1.2643)	(2.2881)
d2_solar_post	-0.0000***		-0.0000	-0.0000***
	(0.0000)		(0.0000)	(0.0000)
dlog_gold	1.5266***	1.5465***	0.1515**	1.4864***
	(0.1037)	(0.1085)	(0.0702)	(0.1019)
d_treasury	0.0851***	0.0890***	0.0756***	0.0810***
	(0.0241)	(0.0259)	(0.0167)	(0.0258)
d_dollar	-0.0090***	-0.0083**	-0.0171***	-0.0079**
	(0.0031)	(0.0033)	(0.0025)	(0.0033)
d2_solar_mw_lag6		-0.0000*		
		(0.0000)		
d2_solar_post_l6		0.0000*		
		(0.0000)		
_cons	0.0076	0.0114	-0.0086	0.0070
	(0.0073)	(0.0075)	(0.0055)	(0.0075)
N	187	181	187	177
R-squared	0.6308	0.6346	0.3451	0.6482

Robust standard errors in parentheses. Copper test: DV = dlog_copper.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Confirming the results, Figure 6 plots actual against fitted values for Model 3, confirming that the model captures the broad directional movements in silver returns throughout the sample period. It also captures deviations concentrated around the 2011 silver price spike and the 2020 COVID period.

Figure 6

7 Conclusion

This paper tests whether the major energy policy commitments of late 2015 and early 2016 generated a structural break in the relationship between silver prices and solar photovoltaic capacity expansion. Using 192 monthly observations spanning January 2010 to December 2025, a Chow test confirms a statistically significant structural break at January 2016 ($F(6,175) = 6.19$, $p < 0.001$). Hypothesis 1, that silver became more broadly responsive to global industrial production after 2016, is not supported. Hypothesis 2, that the solar PV channel specifically drives a structural change in silver prices after January 2016, is supported at the 1% significance level and survives all four robustness checks.

The most important finding is the reversal in the sign of the solar-silver relationship after 2016. Rather than solar deployment driving higher silver prices through a demand channel, the post-2016 period is characterised by a negative relationship between solar capacity growth and silver returns. This is interpreted as evidence of a technology efficiency channel: the same policy commitments that accelerated deployment incentivised manufacturers to reduce silver intensity per panel. Furthermore, this weakened price transmission from solar markets to silver markets. For investors who hold silver as a proxy for energy transition demand growth, this finding is a caution: growing solar deployment does not straightforwardly translate into higher silver prices post-2016, as the efficiency channel partially offsets the demand effect.

Nonetheless, several limitations call for acknowledgement. Mild ARCH effects suggest that another econometric framework, like GARCH, may better capture silver's volatility dynamics. A further limitation is the use of cumulative solar capacity, which is a stock variable and may not fully reflect marginal demand for silver driven by new installations. Therefore, the estimated effect may partly reflect underlying trends rather than demand. Finally, the interaction term framework assumes the structural break is discrete and instantaneous; in practice, market incorporation of policy commitments may be more gradual, suggesting a time-varying coefficient model as a further extension. Future research could extend this framework to other critical minerals to assess whether the efficiency-thrifting dynamic identified here is specific to silver or broader across energy transition commodity markets.

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